

GH Raisonni College of Engineering and Management, Pune

Department of Electronics and Telecommunication Engineering

Course: Digital Logic Design (TAE-2 Project Based Learning)

Topic: Vending Machine Controller Using FSM

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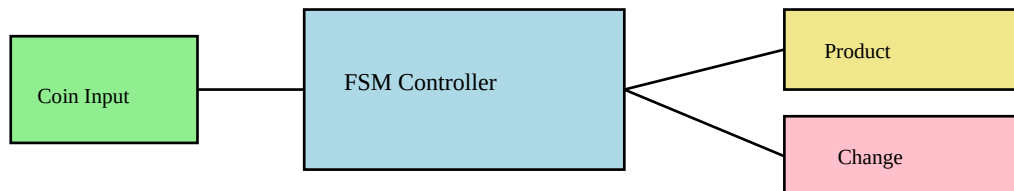
1. Introduction

A vending machine controller is a sequential digital system that accepts ₹5 and ₹10 coins, calculates the balance, dispenses a product costing ₹15, and returns ₹5 change when ₹20 is inserted. The controller is implemented using a Moore Finite State Machine in VHDL.

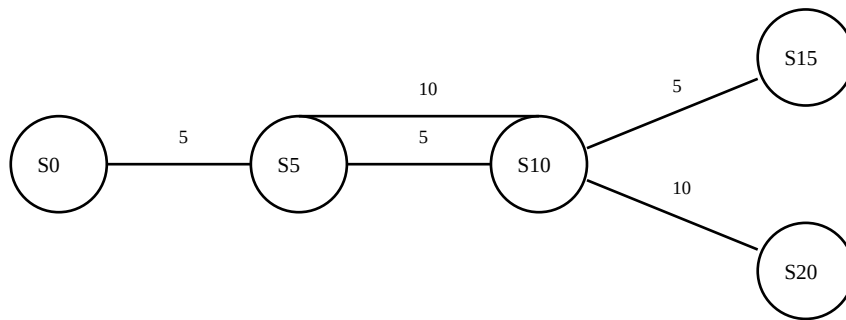
2. Objectives

- Detect ₹5 and ₹10 coins.
- Maintain balance using FSM states.
- Dispense product at ₹15.
- Return ₹5 change when ₹20 is inserted.
- Simulate the design using ModelSim.

3. Block Diagram



4. FSM State Diagram



5. State Table

Present State	Coin=■5	Coin=■10	Dispense	Change
S0	S5	S10	0	0
S5	S10	S15	0	0
S10	S15	S20	0	0
S15	S0	S0	1	0
S20	S0	S0	1	1

6. State Assignment

State	Binary Encoding	Meaning
S0	000	Initial balance ■0
S5	001	Balance ■5
S10	010	Balance ■10
S15	011	Dispense Product
S20	100	Dispense Product + Return ■5 Change

7. State Transitions

On every rising edge of the clock, the current state changes to the next state according to the inserted coin. Reset immediately returns the machine to state S0.

8. Output Logic

Current State	Dispense	Change	Operation
S0	0	0	Waiting for coin
S5	0	0	Balance = ■5
S10	0	0	Balance = ■10
S15	1	0	Dispense product
S20	1	1	Dispense product and return ■5

9. VHDL Code (Summary)

The controller is implemented using two VHDL processes: (1) State Register Process and (2) Next State & Output Logic Process. Five Moore FSM states are declared using an enumerated data type.

10. Testbench

The testbench generates a clock signal and applies the following sequence: Reset → ■5 → ■10 → ■10 → ■10. This verifies product dispensing and change return.

11. Output of VHDL Code

After running the simulation in ModelSim, capture the waveform showing clk, reset, coin5, coin10, dispense, and change signals, and paste it here.

12. Simulation Result

Time Interval	Input	Expected State	Dispense	Change
0–20 ns	Reset	S0	0	0
20–40 ns	■5	S5	0	0
40–60 ns	■10	S15	1	0
80–100 ns	■10	S10	0	0
100–120 ns	■10	S20	1	1

13. Conclusion

The vending machine controller was successfully modeled using a Moore FSM in VHDL. The controller correctly detects ■5 and ■10 coins, dispenses the product when the balance reaches ■15, returns ■5 change for ■20, and resets for the next transaction. The design is verified using ModelSim simulation.